

Delphi

Decision intelligence with an AI second brain

P-ai Project Proposal | Fall 2026

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Overview

Delphi is an ongoing research initiative exploring how AI can extend human cognition and improve decision-making while preserving human agency. We want to build a system that remembers the context behind a person's decisions, helps them compare possibilities, and knows when to challenge their assumptions. The central question is whether persistent memory and thoughtful disagreement can make AI a better thinking partner over time.

For P-ai, we propose a working web prototype that combines a decision workspace with a "second brain" layer. Users will record a question, alternatives, supporting information, and their reasoning. With their permission, Delphi will retrieve relevant notes and past decisions, surface connections, and generate a structured response that identifies tradeoffs, uncertainty, and counterarguments. Users can revisit decisions and record outcomes, creating a feedback loop between what they expected and what happened.

The broader vision includes collective intelligence and forecasting: understanding how people and AI can pool information while preserving independent judgment. This semester will focus on the individual decision and memory workflow, with a small shared forecasting experiment as a stretch goal. Sebastian is pursuing faculty mentorship and grant support, with the goal of connecting Cognitive Science and Computer Science perspectives. Funding and departmental participation are not prerequisites for the prototype.

Goals and semester deliverables

1. Build a usable decision workspace. Support creating, revisiting, and updating decision records, including alternatives, evidence, expectations, and outcomes.
2. Implement user-controlled memory. Store notes and decision history, retrieve relevant context with source references, and let users inspect, correct, and delete saved information.
3. Develop calibrated dissent. Have the system identify unsupported assumptions and offer relevant counterarguments while acknowledging uncertainty. Disagreement should be useful and evidence-based, not automatic.
4. Evaluate the design. Compare a baseline assistant, an assistant with memory, and an assistant with memory plus dissent on the same scenarios. Assess retrieval accuracy, source support, recognition of contradictions, and usefulness of reasoning. Deliver a documented prototype, evaluation results, and a final presentation.

Proposed tech stack

Frontend: React/Next.js with TypeScript for the decision workspace, memory editor, and outcome tracking. Backend: Python with FastAPI for retrieval, model calls, and evaluation. Storage: Supabase/PostgreSQL with pgvector for structured records and semantic retrieval. AI: a hosted LLM API and an embedding model, selected according to available credits and cost. We will use existing models rather than train a foundation model.

Development: GitHub for version control, issues, and code review; pytest for backend and retrieval tests; Figma for interface prototypes. We will keep the architecture small and adjust tooling to the team's experience. Synthetic decision histories and public material will support initial development, avoiding dependence on access to private datasets.

Member requirements and team structure

We propose recruiting 6–8 members in addition to the two co-PMs, with an expected commitment of 4–6 hours per week, including one weekly team meeting and a collaborative work session. Members will work across application development, AI/memory and evaluation, and cognitive science/UX research. Both PMs will coordinate scope, task ownership, integration, and member support.

For engineering roles, introductory experience in Python, JavaScript, or another programming language is helpful; prior machine learning experience is not required. We also welcome students without programming experience to design decision scenarios, research cognitive biases, conduct usability walkthroughs, evaluate responses, and shape the interface. Starter tasks, paired work, and short tutorials will help members learn and move into technical tasks if they wish.

Timeline: September 8 through early December

September 8–20: Onboard members, agree on scope and roles, review relevant research, design sample decision scenarios, and set up the repository and interface sketches.

September 21–October 4: Build the basic decision workspace, database, and baseline assistant. Establish evaluation criteria before comparing features.

October 5–18: Add persistent memory, semantic retrieval, source references, and controls for editing and deleting memories. Test on synthetic histories.

October 19–November 1: Implement calibrated dissent and outcome tracking. Integrate the interface and backend, allowing flexibility around fall break.

November 2–15: Compare the three assistant configurations, run voluntary usability walkthroughs, and fix retrieval, reasoning, and interface problems.

November 16–29: Refine the prototype and document findings, with reduced expectations during Thanksgiving. If the core system is stable, explore a shared forecasting exercise using probabilities and recorded outcomes, without real-money transactions.

November 30–December 6: Freeze features, complete documentation, and prepare the demo and presentation. Finish before finals, adjusting the presentation date to P-ai's schedule.

Responsible design

Memory will be opt-in, private by default, and editable. We will test for fabricated memories, misplaced confidence, and agreement that reinforces bias. The interface will distinguish saved facts from model inferences and keep decisions with the user. Any formal human-subjects research will be scoped with faculty guidance and appropriate review. Longer-term commercialization may follow the research, but this semester's purpose is learning, building, and evaluating the idea.